THE THIRD WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION
March 2, 1940

Morning Session

1. Prove that if $f(x)$ is a polynomial with integral coefficients, and there exists an integer k such that none of the integers $f(1), f(2), \ldots, f(k)$ is divisible by k, then $f(x)$ has no integral root. (page 133)

2. Let A and B be two fixed points on the curve $y = f(x)$, where $f(x)$ is continuous and has a continuous derivative, and the arc AB is concave to the chord AB. If P is a point of the arc AB for which $AP + PB$ is a maximum, prove that PA and PB are equally inclined to the tangent to the curve $y = f(x)$ at the point P. (page 133)

3. Find $f(x)$ such that

$$\int [f(x)]^n \, dx = \left[\int f(x) \, dx\right]^n,$$

when constants of integration are suitably chosen. (page 135)

4. The parabola $y^2 = -4px$ rolls without slipping around the parabola $y^2 = 4px$. Find the equation of the locus of the vertex of the rolling parabola. (page 137)

5. Prove that the simultaneous equations

$$x^4 - x^2 = y^4 - y^2 = z^4 - z^2$$

are satisfied by the points of four straight lines and six ellipses, and by no other points. (page 139)

6. $f(x)$ is a polynomial of degree n, such that a power of $f(x)$ is divisible by a power of its derivative $f'(x)$; i.e., $[f(x)]^p$ is divisible by $[f'(x)]^q$; p, q, positive integers. Prove that $f(x)$ is divisible by $f'(x)$ and that $f(x)$ has a single root of multiplicity n. (page 140)

7. If $u_1^2 + u_2^2 + \cdots$ and $v_1^2 + v_2^2 + \cdots$ are convergent series of real constants, prove that

$$(u_1 - v_1)^p + (u_2 - v_2)^p + \cdots, \; p \text{ an integer} \geq 2,$$

is convergent. (page 141)

8. A triangle is bounded by the lines

$$A_1x + B_1y + C_1 = 0, \qquad A_2x + B_2y + C_2 = 0,$$
$$A_3x + B_3y + C_3 = 0.$$

Show that the area, disregarding sign, is

$$\frac{\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}^2}{2\begin{vmatrix} A_2 & B_2 \\ A_3 & B_3 \end{vmatrix} \cdot \begin{vmatrix} A_3 & B_3 \\ A_1 & B_1 \end{vmatrix} \cdot \begin{vmatrix} A_1 & B_1 \\ A_2 & B_2 \end{vmatrix}}.$$

(page 142)

Afternoon Session

9. A projectile, thrown with initial velocity v_0 in a direction making angle α with the horizontal, is acted on by no force except gravity. Find the length of its path until it strikes a horizontal plane through the starting point. Show that the flight is longest when

$$\sin \alpha \log (\sec \alpha + \tan \alpha) = 1.$$

(page 144)

10. A cylindrical hole of radius r is bored through a cylinder of radius R $(r \leq R)$ so that the axes intersect at right angles.

(i) Show that the area of the larger cylinder which is inside the smaller can be expressed in the form

$$S = 8r^2 \int_0^1 \frac{1 - v^2}{\sqrt{(1 - v^2)(1 - m^2v^2)}}\, dv \quad \text{where} \quad m = \frac{r}{R}.$$

(ii) If

$$K = \int_0^1 \frac{dv}{\sqrt{(1 - v^2)(1 - m^2v^2)}} \quad \text{and} \quad E = \int_0^1 \sqrt{\frac{1 - m^2v^2}{1 - v^2}}\, dv$$

show that

$$S = 8[R^2E - (R^2 - r^2)K].$$

(page 146)

11. From any point (a, b) in the Cartesian plane, show that (i) three normals, real or imaginary, can be drawn to the parabola $y^2 = 4px$; (ii) these

are real and distinct if $4(2p - a)^3 + 27pb^2 < 0$; (iii) two of them coincide if (a, b) lies on the curve $27py^2 = 4(x - 2p)^3$; (iv) all three coincide only if (a, b) is the point $(2p, 0)$. (page 147)

12. Prove that the locus of the point of intersection of three mutually perpendicular planes tangent to the surface

$$ax^2 + by^2 + cz^2 = 1 \qquad (abc \neq 0) \tag{1}$$

is the sphere

$$x^2 + y^2 + z^2 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}. \tag{2}$$

(page 149)

13. Determine all rational values for which a, b, c are the roots of

$$x^3 + ax^2 + bx + c = 0.$$

(page 157)

14. Prove that

$$\begin{vmatrix} a_1^2 + k & a_1a_2 & a_1a_3 & \dots & a_1a_n \\ a_2a_1 & a_2^2 + k & a_2a_3 & \dots & a_2a_n \\ \dots & \dots & \dots & \dots & \dots \\ a_na_1 & a_na_2 & a_na_3 & \dots & a_n^2 + k \end{vmatrix}$$

is divisible by k^{n-1} and find its other factor. (page 158)

15. Which is greater

$$(\sqrt{n})^{\sqrt{n+1}} \quad \text{or} \quad (\sqrt{n+1})^{\sqrt{n}}$$

where $n > 8$? (page 160)